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MOVABLE PLATFORMS FOR VEHICLE CAPTURE ASSEMBLIES AND RELATED DEVICES, ASSEMBLIES, SYSTEMS, AND METHODS

Abstract

Movable platforms for use with a vehicle capture assembly of a capture vehicle and related devices, systems, and methods include floors with rotatable legs extending between the floors. The rotation of the legs relative to one or more of the floors enables the movable platform to move from an initial position to a displaced position as one or more of the floors move relative to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/990,245, filed Nov. 18, 2022, for “MOVABLE PLATFORMS FOR VEHICLE CAPTURE ASSEMBLIES AND RELATED DEVICES, ASSEMBLIES, SYSTEMS, AND METHODS,” the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to systems, devices, assemblies, apparatus, and methods for vehicle (e.g., spacecraft) docking. In some embodiments, the present disclosure includes a probe movement feature, such as, for example, a movable platform, for use with vehicle capture assembly for engaging with an associated vehicle and related devices, systems, and methods.

BACKGROUND

[0003] Docking assemblies and devices may be utilized to mechanically connect two or more vehicles (e.g., spacecraft) to each other. Such spacecraft may be vehicles (e.g., self-propelled vehicles) designed for short-term space flights and/or may be configured to remain in space for a long period of time. The spacecraft may be intended to perform a specific function in a space mission, such as supplying resources to and/or altering the orbit of a target vehicle. In some instances, the spacecraft may be a space station, satellite, or another suitable structure.

[0004] The connection of two or more spacecraft may enable the transfer of resources from one spacecraft to another spacecraft. For example, a spacecraft may dock with a space station to deliver crew and resources. In another example, a spacecraft may dock with a satellite to perform maintenance and repair of one or more components of that satellite. In yet an additional example, a spacecraft may dock with another vehicle to provide a specific mission function, such as, for example, a descent to or an ascent from an astronomical body or to transfer to a select location for the mission.

[0005] Conceptualized methods of docking to spacecraft consist of complex mechanical implements. However, mechanical complexity that is present in many of the above designs, particularly complexity associated with multi-axis servo motor and robotic control systems, increases the likelihood of component failure, which can result in failure in the docking and maintenance process. Further, relative motion between the docked vehicles and applied to the vehicles and the docking assemblies may jeopardize the docking procedure and may also cause damage to one or more of the vehicles and their respective docking assemblies.

[0006] BRIEF SUMMARY

[0007] In some aspects, the disclosure described herein relates to a movable platform for use with a vehicle capture assembly of a capture vehicle, the movable platform including: a first floor configured to be coupled with a portion of the vehicle capture assembly proximate the capture vehicle; a second floor configured to be coupled with a probe assembly of the vehicle capture assembly; and at least one set of legs extending between the first floor and the second floor, the at least one set of legs being rotationally coupled to at least one of the first floor or the second floor and configured to rotate relative to the at least one of the first floor or the second floor; wherein rotation of the at least one set of legs relative to the at least one of the first floor or the second floor enables the movable platform to move from an initial position to a displaced position as the first floor translates laterally relative to the second floor in a direction transverse to a longitudinal axis of the movable platform.

[0008] In some aspects, the disclosure described herein relates to a movable platform for use with a vehicle capture assembly of a capture vehicle, the movable platform including: a first floor configured to be coupled proximate the capture vehicle; a second floor configured to be coupled

with a probe assembly of the vehicle capture assembly; at least one set of legs extending between the first floor and the second floor, the at least one set of legs being rotationally coupled to at least one of the first floor or the second floor and configured to rotate relative to the at least one of the first floor or the second floor, wherein rotation of the at least one set of legs relative to the at least one of the first floor or the second floor enables the movable platform to move from an initial position to a displaced position as the first floor moves relative to the second floor in a lateral direction; and a floating stop configured to rotate when the at least one set of legs rotate in one rotational direction relative to the at least one of the first floor or the second floor.

[0009] In some aspects, the disclosure described herein relates to a vehicle capture system for use with a capture vehicle, the vehicle capture system including: a probe assembly including one or more retention elements for engaging with and securing a target vehicle; and a movable platform coupled to at least a portion of the probe assembly, the movable platform including: a first lateral section coupled with a portion of the capture vehicle; a second lateral section coupled with the probe assembly; and legs extending between the first lateral section and the second lateral section, the legs being rotationally coupled to at least one of the first lateral section or the second lateral section and configured to rotate relative to the at least one of the first lateral section or the second lateral section, wherein rotation of the legs relative to the at least one of the first lateral section or the second lateral section enables the movable platform to move from an initial position to a displaced position as the second lateral section translates relative to the first lateral section.

[0010] In some aspects, the disclosure described herein relates to a method of capturing a spacecraft, the method including: extending a probe of a vehicle capture assembly toward a target spacecraft on a boom, the vehicle capture assembly being coupled to a capture vehicle; enabling motion of the probe relative to the capture vehicle with a movable platform, a first lateral section of the movable platform coupled to the probe and a second lateral section of the movable platform coupled to the boom with rotatable legs extending between the first lateral section and the second lateral section, the enabling motion of the probe relative to the capture vehicle with the movable platform including: translating the first lateral section relative to the second lateral section by rotating the legs relative to the first lateral section and the second lateral section; and while translating the first lateral section relative to the second lateral section, substantially maintaining a rotational position of the probe; and engaging the probe of the vehicle capture assembly with the target spacecraft.

[0011] In some aspects, the disclosure described herein relates to a method of moving a vehicle capture assembly of a capture vehicle with a movable platform, the method including: minimizing deadband in the movable platform by securing the movable platform in an initial position with a biasing mechanism and enabling the movable platform to move to a displaced position when a force threshold is exceeded; and once the force threshold is exceeded, translating a first lateral section of the movable platform coupled to the vehicle capture assembly relative to a second lateral section of the movable platform coupled to the capture vehicle by rotating rotatable legs of the movable platform relative to the first lateral section and the second lateral section, the rotatable legs extending between the first lateral section and the second lateral section.

[0012] The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain

embodiments and do not limit the disclosure.

[0014] FIG. 1 is a schematic side view of a capture vehicle with a vehicle capture assembly and a target vehicle according to one or more embodiments of the disclosure.

[0015] FIG. 2 is an isometric view of a vehicle capture assembly in an initial position according to one or more embodiments of the disclosure.

[0016] FIG. 3 is an isometric view of a movable platform (e.g., a probe movement feature) in an initial position according to one or more embodiments of the disclosure.

[0017] FIGS. 4 and 5 are side views of a movable platform in displaced positions according to one or more embodiments of the disclosure.

[0018] FIGS. 6 and 7 are enlarged views of the movable platform in FIGS. 4 and 5, respectively, in displaced positions.

[0019] FIG. 8 is an isometric view of a movable platform in a displaced position according to one or more embodiments of the disclosure.

[0020] While the disclosure is amenable to various modifications and alternate forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

DETAILED DESCRIPTION

[0021] As used herein, the terms “substantially,” “about,” and “approximate,” in reference to a given parameter, means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially, about, or approximately met may be at least about 90% met, at least about 95% met, at least about 99% met, or even 100% met.

[0022] Embodiments of the present disclosure may include movable platforms for use with capture and mating assemblies and systems used for mechanical docking of two or more vehicles (e.g., spacecraft in orbit, with or without self-propulsion), using one or more vehicle capture device or assembly on the capture vehicle. Such vehicle capture assemblies are configured to engage with a docking portion of a target vehicle to be captured (e.g., one or more docking cones coupled to the target vehicle). Some embodiments may enable the autonomous capture and docking of spacecraft with relatively large mass and inertia, while incurring minimal disturbance to either vehicle. Some embodiments provide benefits in the form of a relatively simple and compact docking architecture with mechanical compliance for improved reliability and safety, that is, preventing damage to the spacecraft.

[0023] Some embodiments of the present disclosure may include a movable platform having movable joints or unions (e.g., rotational joints) enabling the docking assembly to move relative to a capture vehicle on which the docking assembly is mounted. For example, the movable platform may include multiple sections (e.g., floors, lateral members, plates) that move relative to each other (e.g., lateral or radial movement in a direction that is transverse to a longitudinal axis of the assembly) of the docking assembly. Such a configuration may enable the lateral movement of the floors through rotation of the legs while keeping the floors substantially parallel to each other. In some embodiments, the rotation of the legs between multiple floors may act as a parallelogram-type movement (e.g., a double parallelogram-type movement) where each set of legs with opposing floors on either side of the legs defines a substantial parallelogram shape. Stated in another way, the rotation of the legs between multiple floors may move enable lateral movement between the floors while maintaining the floors in a substantially parallel orientation.

[0024] In some embodiments, the movement of the legs may minimize (e.g., substantially prevent) rotation of the floors of the docking assembly relative to each. For example, such a configuration may substantially prevent rotation of the floors relative to each other with the legs or unions

between the floors while enabling the lateral movement of the floors through rotation of the legs and keeping the floors substantially parallel to each other.

[0025] Embodiments of the disclosure enable angular displacement of one or more sets of legs that, in turn, enable lateral displacement of the floors relative to each other. At least some of the floors may be attached to components of a vehicle capture system. For example, the floors may be attached, on one end of the movable platform, to a docking assembly and, on another end of the movable platform, to spacecraft (e.g., a chaser or capture vehicle).

[0026] Embodiments of the movable platform may include a biasing mechanism (e.g., springs, such as torsion springs). For example, the biasing mechanism may resist deflection for loads that are below a threshold level (e.g., about 1 N, 2 N, or more or less depending on the application) while enabling deflection once the threshold level is overcome. Once the threshold level is overcome, additional force may be required to continue moving the movable platform. For example, the deflection may increase as the loading (e.g., lateral loading) increases. The restorative nature of the biasing mechanism may assist to minimize any offset between the capture vehicle and the target vehicle during a capture and docking operation as the biasing mechanism tends to force the movable platform back to the initial position.

[0027] In some embodiments, the biasing mechanism may include a floating stop that acts to center the movable platform in an initial position and to force the movable platform back toward the initial position after a displacement. The use of the biasing mechanism including the floating stop may minimize or even eliminate deadband (e.g., play or slop) during movement of the movable platform between an initial position and a displaced position. For example, the biasing force acting on the floors via the legs may act to secure the flooring in the initial position until the threshold force is overcome enabling the floors to move while reducing or eliminating any unwanted movement (e.g., play) between the floors.

[0028] Embodiments of the movable platform may include linkages or legs (e.g., pairs of linkages or legs extending between floors). The legs rotate relative to pinned joints on each floor, enabling the floors to move relative to one another. Embodiments of the movable platform may utilize redundant revolute joints at the pinned linkage joints in order to reduce fault caused by friction as compared to alternative compliance devices. For example, if one revolute joint is to fail (e.g., via binding, galling, etc.) the other joint may continue to operate under substantially normal conditions. In some embodiments, further reduction in friction, if desired, may be enabled through the use of rolling element bearings (e.g., ball bearings). A centering mechanism comprised of redundant torsion springs and one or more floating stops allows the mechanism to return to the nominal, centered position after an initial displacement of one or both of the movable platform floors. Preloads in the torsion springs also allow the mechanism to resist motion (e.g., substantially prevent) for loads less than a specific threshold.

[0029] In operation in space, a target vehicle and chaser vehicle generally cannot be perfectly aligned via rendezvous and proximity operations prior to docking operations. Compliance features in the docking mechanism may enable the successful completion of the docking operation and accommodate misalignments between the two vehicles. Embodiments of the movable platform may provide lateral compliance between the target vehicle and chaser vehicle in a relatively simple and compact manner by enabling lateral displacements of the docking mechanism relative to the chaser vehicle while reducing disturbances to the vehicles. The centering biasing mechanism may assist in removing misalignment between the two vehicles during the docking operation. Embodiments of the movable platform may be unique in that the movable platform utilizes redundant revolute joints, which enables greater fault tolerance (e.g., to friction-type failures) as compared to many similar solutions that tend to bind during use. Further, the movable platform may provide a compliance feature that may be packaged into a relatively shorter and more compact length along the boom as compared to other solutions.

[0030] In some embodiments, the biasing mechanism of the movable platform may act as

attenuation elements or features to assist in dampening movement of the docking assembly (e.g., with the target vehicle at least partially coupled thereto) relative to the capture vehicle. Such docking assemblies may provide initial compliance for soft capture and/or may at least partially attenuate relative motion between the vehicles (e.g., before retraction of the docking assembly to a final rigidized connection).

[0031] FIG. 1 depicts a schematic side view in which a capture vehicle **10** (e.g., a servicing vehicle, a chaser spacecraft, a transfer spacecraft, etc.) may be operated to approach, capture, dock to, supply, transfer cargo or resources to, and/or service a target vehicle **11**, according to one or more embodiments of the disclosure.

[0032] Capture vehicle **10** and target vehicle **11** each may be a spacecraft or a satellite situated in orbit around a body. The capture vehicle **10** may be a spacecraft designed to approach, capture, dock to, and undock from the target vehicle **11**. Docking of the capture vehicle **10** to target vehicle **11** may enable a specific function in a space mission. For example, the connection of the vehicles **10**, **11** may enable the transfer of resources (e.g., cargo, equipment, passengers, crew, etc.) from one vehicle to another vehicle, may enable vehicle repair, and/or may enable a specific mission function (e.g., a descent to or an ascent from an astronomical body or to transfer to a select location in space for the mission).

[0033] Capture vehicle **10** may be designed to dock with more than one target vehicle **11**. For example, the capture vehicle **10** may be provided with a docking mechanism (e.g., vehicle capture assembly **22**) that enables the capture vehicle **10** to dock and undock from multiple target vehicles **11**. The capture vehicle **10** may be configured to dock with one or more of the target vehicles **11** comprising one or more docking elements **18** (e.g., a docking cone, an engine, etc.).

[0034] As discussed below in greater detail, the vehicle capture assembly **22** may include a movable platform **24** including one or more attenuation elements or features to assist in enabling movement of the vehicle capture assembly **22** relative to the capture vehicle **10** while also dampening such movement of the vehicle capture assembly **22** (e.g., with the target vehicle **11** at least partially coupled thereto) relative to the capture vehicle **10**. The vehicle capture assembly **22** and the movable platform **24** may provide initial compliance for soft capture of the target vehicle **11** and/or may at least partially attenuate relative motion between the vehicles **10**, **11** (e.g., before retraction of the vehicle capture assembly **22** to a final rigidized connection).

[0035] In some embodiments, the movable platform **24** may be passive. For example, the movable platform **24** may function in a manner that does not require the use of active components such as a motor directly actuating the movable platform **24** (e.g., is not required to be driven in an active manner). The movable platform **24** may use passive methodology or mechanisms, such as mechanical forces (e.g., biasing members, dampening mechanisms, etc.) to enable movement of the vehicle capture assembly **22**. In additional embodiments, the movable platform **24** may include active (e.g., actively driven) components enabling and/or regulating motion in one or more of the degrees of freedom.

[0036] As depicted, the capture vehicle **10** may include a spacecraft body **12**, a docking platform **14**, a main thruster **17**, gimbaled thrusters **20**, and the vehicle capture assembly **22**. As noted above, the vehicle capture assembly **22** may include retention elements that directly contact and secure the target vehicle **11** in a manner that does not require the use of active components such as a motor directly actuating the retention elements. Rather, the retention elements may use passive methodology or mechanisms, such as mechanical forces (e.g., biasing forces), to engage with the target vehicle **11**.

[0037] As discussed below, while a motor (e.g., only a single and solitary motor) may be used to actively move (e.g., translate) the vehicle capture assembly **22** toward and/or away from the target vehicle **11**, such a motor may only indirectly contribute to the engagement and/or disengagement of the retention elements. For example, while the motor may place the retention elements in a selected position relative to the target vehicle, a force applied to the vehicle capture assembly **22** may be

utilized to engage and/or disengage the retention elements (e.g., a force overcoming one or more biasing elements of the vehicle capture assembly **22**) in a passive manner that is not actively driven by a motor or an otherwise electronic device.

[0038] Target vehicle **11** may be a spacecraft to be captured by the vehicle capture assembly **22** of the capture vehicle **10**. Target vehicle **11** may be in low earth orbit, medium earth orbit, geosynchronous orbit, beyond geosynchronous orbit, or in another orbit around an astronomical body, for example, such as Earth, a moon, or another planetary body. Target vehicle **11** may include the docking element **18** and a separation ring **19**.

[0039] Vehicle capture assembly **22** of capture vehicle **10** may be configured to capture target vehicle **11** at docking element **18** and to pull target vehicle **11** and capture vehicle **10** together for docking. When docked, one or more portions of the target vehicle **11** and/or vehicle capture assembly **22** may abut and retain the vehicles **10**, **11** together.

[0040] FIG. **2** depicts an isometric view of a vehicle capture assembly **100** in an initial position that may be used with a capture vehicle. In some embodiments, vehicle capture assembly **100** may be similar to, and include similar components and features of, the vehicle capture assembly **22** of the capture vehicle **10** which is depicted schematically in FIG. **1** and discussed above. In some embodiments, the vehicle capture assembly may be similar to that disclosed in U.S. patent application Ser. No. 17/207,646, filed Mar. 20, 2021, published as US 2021/0339893 A1, and/or U.S. patent application Ser. No. 15/829,807, filed Dec. 10, 2017, issued as U.S. Pat. No.

11,104,459, the disclosure of each of which is incorporated herein in its entirety by reference.

[0041] As shown in FIG. **2**, the vehicle capture assembly **100** includes a probe or probe assembly **102** coupled to a lance or lance assembly **104** (e.g., coupled via a movable platform **101** at a distal portion or end of the lance assembly **104**). The probe assembly **102** includes one or more retention features (e.g., barbs **106**) that extend from the probe assembly **102** at a location proximate a probe tip **108**. The barbs **106** may extend in a direction transverse to a length or longitudinal axis of one or more portions of the vehicle capture assembly **100** (e.g., lateral to a length of the lance assembly **104**). As depicted, the rotatable barbs **106** extend laterally outward and in a proximal direction toward the lance assembly **104** in order to capture a target vehicle **11** (FIG. **1**).

[0042] In some embodiments, the barbs **106** may be biased (e.g., spring-loaded) in a selected position. For example, the barbs **106** may be in the depicted deployed position where the barbs **106** may couple with a portion of the target vehicle **11** (FIG. **1**). In additional embodiments, the barbs **106** may be biased in a retracted or stowed position.

[0043] Referring to FIGS. **1** and **2**, the lance assembly **104** may include functionality that enables extension and/or retraction of a portion of the probe assembly **102** to facilitate docking of capture vehicle **10** with target vehicle **11**. For example, when capture vehicle **10** is positioned proximate to the target vehicle **11**, the probe assembly **102** may be extended to and inserted into the docking element **18** of the target vehicle **11** with the lance assembly **104**. The lance assembly **104** may include a lance boom **110** that is driven by a motor **112** of the probe assembly **102**. The motor **112** may be used to actively move (e.g., translate) the lance boom **110** toward and/or away from the target vehicle **11**.

[0044] In some embodiments, the motor **112** may only indirectly contribute to the engagement and/or disengagement of the barbs **106**. For example, while the motor **112** may place the barbs **106** in a selected position relative to the target vehicle **11**, force applied to the barbs **106** (e.g., to overcome the biasing force of the barbs **106** into the deployed position) may be applied as the barbs **106** are inserted into the docking element **18** to engage the barbs **106** in a passive manner that is not actively driven by the motor **112**. As discussed below, movement of the probe assembly **102** (e.g., by forcing the probe assembly **102** into the target vehicle **11**) may be used to release the barbs **106** from the target vehicle **11** (e.g., by overcoming the biasing forces of the barbs **106** in a different manner with internal components of the probe assembly **102**).

[0045] The vehicle capture assembly **100** may include a backstop plate **118** for mating with a

portion of the target vehicle **11** (e.g., the docking element **18** in the captured position).

[0046] As depicted, the vehicle capture assembly **100** includes a probe movement feature (e.g., a movable platform **101**). In some embodiments, the movable platform **101** may be similar to the movable platform **24** discussed above.

[0047] As depicted, the movable platform **101** may include floors at opposing ends of the movable platform **101** (e.g., a first end floor **120** and a second end floor **122**). For example, the first end floor **120** may be coupled (e.g., rigidly coupled in a substantially immovable manner) to the capture vehicle **10** (FIG. 1). The second end floor **122** may be coupled (e.g., rigidly coupled) to the vehicle capture assembly **100** (e.g., to the probe assembly **102**).

[0048] The second end floor **122** may be movably coupled to the first end floor **120** via one or more movable unions or joints to enable the second end floor **122** to move relative to both the first end floor **120** and the capture vehicle **10**. Each of the movable unions (e.g., linkages, legs **126**, etc.) may provide at least one degree of freedom of movement between the second end floor **122** and the first end floor **120**. As discussed above, the movable unions may limit movement between the second end floor **122** and the first end floor **120** to primarily translational movement (e.g., lateral, side-to-side translation in a direction transverse to a direction of movement of the lance assembly **104**) while keeping the second end floor **122** and the first end floor **120** in an orientation substantially parallel to each other.

[0049] As depicted, an intermediate floor **124** may be positioned between the first end floor **120** and the second end floor **122**. Sets of legs **126** may extend between each of the floors **120**, **122**, **124** and the legs **126** may rotate relative to two of the respective floors **120**, **122**, **124** to which each leg **126** is connected (e.g., via a rotational joint, such as a pinned union). Rotation of the legs **126** may respectively enable each of the floors **120**, **122**, **124** to move laterally relative to each other in a lateral direction or axis and/or radial direction or axis that is transverse (e.g., substantially perpendicular) to a longitudinal axis of the lance assembly **104** along which the lance boom **110** moves.

[0050] In some embodiments, the movable platform **101** may include central openings to accommodate a central lanyard **128** (e.g., housed within a flexible conduit **130**) extending through the movable platform **101**. The central lanyard **128** and accompanying flexible conduit **130** may both be bendable to accommodate movement of the floors **120**, **122**, **124** of the movable platform **101**. The central lanyard **128** may be coupled to and act to move the barbs **106**. For example, the central lanyard **128** may pull on the barbs **106** to retract the barbs **106** and release the target vehicle **11**.

[0051] FIG. 3 is an isometric view of a movable platform **200**. In some embodiments, the movable platform **200** may be similar to the movable platforms **24**, **122** discussed above.

[0052] As depicted, the movable platform **200** may include end members, sections, or plates (e.g., a first floor **202** and a second floor **204**) where one of the first floor **202** or the second floor **204** may be coupled (e.g., rigidly coupled) to the capture vehicle **10** (FIG. 1) while the other is coupled to at least a portion of the vehicle capture assembly **100** (FIG. 2).

[0053] The second floor **204** may be movable (e.g., primary translatable) relative to the first floor **202** via one or more movable unions or legs **208** to enable the second floor **204** to move relative to both the first floor **202** and the capture vehicle **10** in a direction transverse to (e.g., substantially perpendicular to) a longitudinal axis **201** or centerline of the movable platform **200**. The movable unions or legs **208** may provide at least one degree of freedom of movement (e.g., substantially lateral translational movement) between the second floor **204** and the first floor **202**. For example, the movable platform **200** may include the first floor **202** coupled to the second floor **204** via the legs **208** to enable the vehicle capture assembly **100** to translate along a lateral axis of movement as the floors **202**, **204** move laterally relative each other.

[0054] As depicted, an intermediate structure or central member (e.g., at least one intermediate floor **206**) may be positioned between the first floor **202** and the second floor **204**. Legs **208** may

extend between each of the floors **202**, **204**, **206**. For example, a first set of the legs **208** (e.g., two radially opposing legs **208**) may extend between the first floor **202** and the intermediate floor **206** and a second set of the legs **208** may extend between the second floor **204** and the intermediate floor **206**. The first set of the legs **208** may be offset from the second set of the legs **208** (e.g., offset about 180 degrees about the longitudinal axis **201**). As discussed below, in some embodiments, the first and second sets of legs **208** may partially overlap along the longitudinal axis **201** such that the first and second sets of legs **208** coextend along a portion of the longitudinal axis **201** while being coupled at opposite ends of the intermediate floor **206** to define an at least partially nested structure.

[0055] The legs **208** may rotate relative to two of the respective floors **202**, **204**, **206** to which each leg **208** is connected (e.g., via a rotational joint, such as a pinned union). Rotation of the legs **208** may respectively enable each of the floors **202**, **204**, **206** to move in a lateral direction that is transverse (e.g., substantially perpendicular) to the longitudinal axis **201**. Floors **202**, **204** may include brackets **210** extending from one side of the respective floor **202**, **204**. Intermediate floor **206** may include brackets **212** extending from both sides of a central structure **214** of the intermediate floor **206**. The brackets **212** and the legs **208** may be configured such that the legs **208** may respectively couple with brackets **212** positioned relatively further away from the brackets **210** on the other floor **202**, **204** to which the leg **208** is coupled. For example, a central portion of the legs **208** (e.g., central rib **242**) may be rounded or curved to pass around the central structure **214** of the intermediate floor **206** to couple with a distal set of brackets **212**. As noted above, such a configuration enable the legs **208** to overlap along the longitudinal axis **201** to define a nested structure (e.g., a relatively more compact structure).

[0056] The movable platform **200** may include one or more biasing mechanisms **218** to bias the movable platform **200** in an initial position (e.g., as shown in FIG. 3). As depicted, the biasing mechanism **218** may comprise torsion springs **220** positioned between one or more of the legs **208** and a respective one of the floors **202**, **204**, **206**. While the biasing mechanism **218** is discussed herein primarily as torsion springs **220**, in additional embodiments, other biasing mechanisms may be implemented (e.g., dampening mechanisms, differing spring configurations, etc.).

[0057] In some embodiments, the biasing mechanism **218** may include a set of torsion springs **220** positioned at an interface or union between a leg **208** and respective bracket **210** of the floors **202**, **204**. As depicted, only one leg **208** of each set may include the torsion springs **220** positioned at the outer end floors **202**, **204**. In additional embodiments, the torsion springs **220** may be positioned on one or more of the legs **208** at the union with the intermediate floor **206** (e.g., as shown in FIG. 2).

[0058] As discussed above, the torsion springs **220** (e.g., along with a floating stop **222**, which is described in detail below) may bias the movable platform **200** in the initial position as depicted in FIG. 3. The torsion springs **220** may act to return the movable platform **200** substantially to the initial position after a force applied to the movable platform **200** causes the legs **208** to rotate in order displace to one or more of floors **202**, **204**, **206**. As also discussed above, the torsion springs **220** may secure the floors **202**, **204**, **206** until a threshold force (e.g., 1N, 2N, etc.) is applied to or between one or more of the floors **202**, **204**, **206**. Once the threshold force is exceeded, one or more sets of the legs **208** may begin to rotate enabling one or more of the floors **202**, **204**, **206** to move relative to each other. In such an embodiment, the torsion springs **220** (e.g., along with the floating stop **222**) may substantially prevent any movement (e.g., slop) in the floors **202**, **204**, **206** until the threshold force is applied. For example, the torsion springs **220** and/or the floating stop **222** may act to minimize or eliminate any deadband in the movement of the floors **202**, **204**, **206**. Once the threshold force is applied, the legs **208** may enable translation of one or more of floors **202**, **204**, **206** to move.

[0059] In some embodiments, rotation of the legs **208** may be limited by a hardstop provided by the movable platform **200**. For example, a portion of the intermediate floor **206** (e.g., cutout portion **240**) may contact the central rib **242** of the legs **208** (e.g., between bracket portions **244** at either

end of the leg **208**) to provide a hardstop for rotation of the legs **208** and translation of one or more of the floors **202**, **204**, **206**.

[0060] As noted above, in some embodiments, the biasing mechanisms disclosed herein may act to attenuate movement during a docking procedure. For example, the biasing features may dampen transitional forces between the vehicles **10**, **11** (FIG. **1**) as the docking assembly **100** is inserted in a docking cone of the target vehicle **11**.

[0061] FIGS. **4** and **5** are side views of a movable platform (e.g., movable platform **200**) being displaced. For example, FIG. **4** shows a set of the legs **208** (e.g., the legs **208** extending between the first floor **202** and the intermediate floor **206**) rotating in a clockwise direction (e.g., about 30 degrees in the clockwise direction as depicted in FIG. **4**). For clarity, only one set of the legs **208** is shown as rotating; however, the other set may also rotate in a similar manner in either direction (e.g., as shown in FIG. **8**). FIG. **5** shows the set of the legs **208** (rotating in a counterclockwise direction (e.g., about 30 degrees in the counterclockwise direction as depicted in FIG. **5**). In additional embodiments, the legs **208** may rotate more or less than 30 degrees in either direction (e.g., ± 15 , 45, 60, 75, 90 degrees or variations in between).

[0062] FIGS. **6** and **7** are enlarged views of the movable platform **200** in the positions shown in FIGS. **4** and **5**, respectively. FIGS. **6** and **7** are enlarged for clarity to describe the function of a floating stop **222** that may be utilized in unison with (e.g., may be part of) the biasing mechanism **218** in order to minimize any deadband in the movement of the movable platform **200**, as is discussed above. Similar to the legs **208**, the floating stop **222** may be rotatably coupled to a respective bracket (e.g., bracket **210** via a pinned connection **216**).

[0063] Referring to FIGS. **4** and **6**, as depicted, rotation of the lower set of legs **208** translates both the second floor **204** and the intermediate floor **206** relative to the first floor **202** in a lateral direction that is transverse to (e.g., set across) the longitudinal axis **201**. Due to the rotation of the legs **208**, the second floor **204** and the intermediate floor **206** will also move along the longitudinal axis **201** (e.g., to move relatively closer or further away from each other). However, as discussed herein, the lateral movement between the floors **202**, **204**, **206** may be the primary type and/or direction of movement for the movable platform **200** (e.g., constituting a majority of the displacement). In use, the second floor **204** and the intermediate floor **206** translate relative to the first floor **202**, which may be coupled to the capture vehicle **10** (FIG. **2**), in order to move the second floor **204**, which is coupled to a probe assembly **102** (FIG. **2**) to move transversely and/or laterally relative to the target spacecraft **11** (FIG. **2**) during a docking procedure. Rotation of the upper set of legs **208** (e.g., into or out of the page as depicted in FIG. **4**) may further translate the second floor **204** relative to the first floor **202** and the intermediate floor **206** (e.g., as shown in FIG. **8**). After displacement, the biasing mechanism **218** may act to force the second floor **204** and the intermediate floor **206** back to being substantially aligned with the longitudinal axis **201** in the initial position. Such alignment may act to align the movable platform **200** with a portion of the target vehicle **11** (e.g., with a centerline of the docking element **18** of the target vehicle **11**).

[0064] During the rotation of the legs **208**, the floating stop **222** may remain in an initial position (e.g., remain stationary) where a stop portion **224** of the floating stop **222** is engaged with a fixed surface of the movable platform **200** (e.g., a fixed portion **226** of one of the brackets **210** creating a hardstop for the floating stop **222** in one direction). Stated in another way, in the movement shown in FIGS. **4** and **6**, the floating stop **222** is substantially fixed (e.g., the floating stop **222** does not substantially rotate as the floating stop **222** is secured by interaction between the stop portion **224** and fixed portion **226** of the bracket **210**). The leg **208** may apply a force against one or more of the torsion springs **220** (e.g., by twisting torsion spring **228**) to overcome the biasing force of the torsion spring **228**. As depicted, a first set of ends of the torsion spring **228** may be secured by a first movable bracket portion **230** of the floating stop **222**. A second set of end of the torsion spring **228** may be secured to the leg **208** (e.g., at leg bracket **234**).

[0065] As depicted, a portion of the floating stop **222** (e.g., a second movable bracket portion **246**)

may act as a hardstop to the leg **208**.

[0066] Once the force is removed, the torsion spring **228** may return the leg **208** to the initial position.

[0067] Referring to FIGS. **5** and **7**, as depicted, rotation of lower set of legs **208** translates both the second floor **204** and the intermediate floor **206** relative to the first floor **202** in a counterclockwise direction, similar to the reverse movement in FIGS. **4** and **6** discussed above.

[0068] During the rotation of the legs **208** in the counterclockwise direction shown in FIGS. **5** and **7**, the floating stop **222** may also rotate in a counterclockwise direction causing the stop portion **224** to move away from the fixed portion **226** of one of the brackets **210**. In some embodiments, a hardstop **236** on the leg **208** may contact the stop portion **224** of the floating stop **222** to rotate the floating stop **222**.

[0069] The leg **208** may apply a force against one or more of the torsion springs **220**. For example, force applied to the leg **208** may rotate the floating stop **222** and, in turn, twist torsion spring **232**, to overcome the biasing force of the torsion spring **232**. As depicted, ends of the torsion spring **232** may be secured by the second movable bracket portion **246** of the floating stop **222** (also shown in FIGS. **5** and **6**). A second set of ends of the torsion spring **232** may be secured to the floor **202** (e.g., at floor bracket **238**).

[0070] When the floating stop **222** rotates a selected amount (e.g., when a portion of the first movable bracket portion **230** contacts the first floor **202**), the floating stop **222** acts as a hardstop to the leg **208**.

[0071] Referring back to FIG. **4**, in some embodiments, similar to the above, rotation of the legs **208** may be limited by one or more of the floating stop **222** or the hardstop provided by the movable platform **200**. For example, the cutout portion **240** of the intermediate floor **206** may contact the central rib **242** of the leg **208** to provide the hardstop for rotation of the leg **208** and translation of one or more of the floors **202**, **204**, **206**.

[0072] As above, once the force is removed, the torsion spring **232** may return the leg **208** to the initial position.

[0073] As also discussed above, the floating stop **222** may enable a configuration with substantially little to no backlash when moving between the initial and displaced positions.

[0074] FIG. **8** is an isometric view of a movable platform **300** shown with multiple sets of legs **302** rotating the movable platform **300** in a displaced position. In some embodiments, the movable platform **300** may be similar to the movable platforms **24**, **122**, **200** discussed above.

[0075] As depicted, each of floors **304** may move relative to each other along one or more translational degrees of freedom (e.g., along the x-axis, the y-axis, and/or the z-axis) in order to enable the opposing end floors to move laterally relative to each other and then to be biased back into an initial position where the floors **304** are substantially laterally aligned relative to each other. The rotation of the legs **302** between each of the floors **304** may act as a double parallelogram-type movement where each set of legs **302** with opposing floors **304** on either side of the legs **302** define a substantial parallelogram shape. For example, the rotation of the legs **302** between the floors **304** may move the floors **304** relative to each other while maintaining the floors **304** in a substantially parallel orientation and without substantially rotating the floors **304** (e.g., changing the rotational position and/or orientation of the floors **304**).

[0076] In some embodiments, the movable platform **300** may include one or more electrostatic discharge (ESD) features, such as, for example, ground wires **306** or a braid may extend between the legs **302** and the floors **304**.

[0077] While certain embodiments discussed herein are directed to the movable platforms having one or more degrees of translational freedom (e.g., three degrees), other embodiments may include other variations of degrees of freedom (e.g., even with some rotational freedom if desired in a select application).

[0078] The embodiments of the disclosure described above and illustrated in the accompanying

drawing figures do not limit the scope of the disclosure, since these embodiments are merely examples of embodiments of the disclosure. The disclosure is defined by the appended claims and their legal equivalents. Any equivalent embodiments lie within the scope of this disclosure. Indeed, various modifications of the present disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, will become apparent to those of ordinary skill in the art from the description. Such modifications and embodiments also fall within the scope of the appended claims and their legal equivalents. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A vehicle capture system for use with a capture vehicle, the vehicle capture system comprising: a probe assembly including one or more retention elements for engaging with and securing a target vehicle; and a movable platform coupled to at least a portion of the probe assembly, the movable platform comprising: a first lateral section coupled with a portion of the capture vehicle; a second lateral section coupled with the probe assembly; and legs extending between the first lateral section and the second lateral section, the legs being rotationally coupled to at least one of the first lateral section or the second lateral section and configured to rotate relative to the at least one of the first lateral section or the second lateral section, wherein rotation of the legs relative to the at least one of the first lateral section or the second lateral section enables the movable platform to move from an initial position to a displaced position as the second lateral section translates relative to the first lateral section.
2. The vehicle capture system of claim 1, further comprising a floating stop configured to rotate when the legs rotate in one rotational direction relative to the at least one of the first lateral section or the second lateral section.
3. The vehicle capture system of claim 2, wherein the floating stop is configured to remain stationary when the legs rotate in another rotational direction relative to the at least one of the first lateral section or the second lateral section, the another rotational direction opposing the one rotational direction.
4. The vehicle capture system of claim 1, further comprising a biasing mechanism configured to bias the movable platform in the initial position.
5. The vehicle capture system of claim 1, wherein the legs exhibit a nested configuration where a first set of the legs longitudinally overlaps with a second set of the legs.
6. The vehicle capture system of claim 5, further comprising a floating stop configured to rotate when the legs rotate in one rotational direction relative to the at least one of the first lateral section or the second lateral section.
7. The vehicle capture system of claim 6, wherein the floating stop is configured to remain stationary when the legs rotate in another rotational direction relative to the at least one of the first lateral section or the second lateral section, the another rotational direction opposing the one rotational direction.
8. The vehicle capture system of claim 5, further comprising a biasing mechanism configured to bias the movable platform in the initial position.
9. A method of capturing a spacecraft, the method comprising: extending a probe of a vehicle capture assembly toward a target spacecraft on a boom, the vehicle capture assembly being coupled to a capture vehicle; enabling motion of the probe relative to the capture vehicle with a movable platform, a first lateral section of the movable platform coupled to the probe and a second lateral section of the movable platform coupled to the boom with rotatable legs extending between the first lateral section and the second lateral section, the enabling motion of the probe relative to the

capture vehicle with the movable platform comprising: translating the first lateral section relative to the second lateral section by rotating the legs relative to the first lateral section and the second lateral section; and while translating the first lateral section relative to the second lateral section, substantially maintaining a rotational position of the probe; and engaging the probe of the vehicle capture assembly with the target spacecraft.

10. The method of claim 9, further comprising, after translating the first lateral section relative to the second lateral section, returning the movable platform to an initial position with a biasing mechanism.

11. The method of claim 10, further comprising minimizing deadband in the movable platform by securing the movable platform in the initial position with the biasing mechanism and enabling the movable platform to move to a displaced position when a force threshold is exceeded.

12. The method of claim 9, further comprising translating the first lateral section relative to the second lateral section while translating the first lateral section and the second lateral section relative to a third intermediate lateral section positioned between the first lateral section and the second lateral section.

13. A method of moving a vehicle capture assembly of a capture vehicle with a movable platform, the method comprising: minimizing deadband in the movable platform by securing the movable platform in an initial position with a biasing mechanism and enabling the movable platform to move to a displaced position when a force threshold is exceeded; and once the force threshold is exceeded, translating a first lateral section of the movable platform coupled to the vehicle capture assembly relative to a second lateral section of the movable platform coupled to the capture vehicle by rotating rotatable legs of the movable platform relative to the first lateral section and the second lateral section, the rotatable legs extending between the first lateral section and the second lateral section.

14. The method of claim 13, further comprising, in a first rotational direction, rotating a floating stop positioned at a union between the rotatable legs and at least one of the first lateral section or the second lateral section against one or more biasing elements.

15. The method of claim 14, further comprising, in a second rotational direction, substantially maintaining the floating stop in a stationary position when the rotatable legs rotate against the one or more biasing elements, the second rotational direction opposing the first rotational direction.

16. The method of claim 14, further comprising, while translating the first lateral section relative to the second lateral section, substantially maintaining a rotational position of the first lateral section relative to the second lateral section.
